

Indian Statistical Institute, Bangalore Centre
B.Math. (III Year): 2008-2009
Semester I: Semestral Examination
Introduction to Differential Equations

11.12.2008

Time : 3 hrs.

Max.Marks:100

Note: Each question carries 16 marks. Any score above 100 will be treated as 100.

1. Let $x(\cdot)$ denote the general solution to

$$x''(t) + px'(t) + qx(t) = 0, \quad t > 0,$$

where p, q are constants. Show that $\lim_{t \rightarrow \infty} x(t) = 0$ if and only if $p > 0, q > 0$.

2. (i) Find the general solution to the problem

$$tx''(t) + ax'(t) = 0, \quad t > 1 \quad \text{with} \quad x(1) = 1,$$

where $a \geq 0$ is a constant.

(ii) For the problem in (i) above, show that a unique bounded solution exists if and only if $a \leq 1$.

3. Find two linearly independent Frobenius series solutions to the ODE

$$2tx''(t) + (3 - t)x'(t) - x(t) = 0, \quad t > 0.$$

4. Find $u(\cdot, \cdot)$ satisfying

$$u_t(t, x) + 2u_x(t, x) + 3u(t, x) = 0, \quad t > 0, x \in \mathbb{R} \quad \text{with} \quad u(0, x) = x^2.$$

5. Find the solution to the Dirichlet problem:

$$\begin{aligned} \Delta u(r, \varphi) &= 0, \quad 1 < r < 2, \quad -\pi < \varphi < \pi, \quad \text{with} \\ u(1, \varphi) &= 1 + \cos^2 \varphi, \quad u(2, \varphi) = \sin^2 \varphi, \quad -\pi \leq \varphi \leq \pi. \end{aligned}$$

Verify that the expression you get is indeed harmonic in the annulus.

6. Find $(t, x) \mapsto u(t, x)$ which is bounded in x and satisfies

$$u_t(t, x) = u_{xx}(t, x) + 3t^2, \quad t > 0, x \in \mathbb{R}$$

with $u(0, x) = \sin x$, $x \in \mathbb{R}$. You may assume facts about characteristic functions of normal distributions. (Hint: Show first that it is enough to find v, w such that v is the bounded solution to the corresponding homogeneous equation with the given initial condition, and w is a particular solution to the given inhomogeneous equation with zero initial condition.)

7. Find u satisfying

$$u_{tt}(t, x) = u_{xx}(t, x) + xt, \quad t > 0, x \in \mathbb{R}$$

with $u(0, x) = x^2$, $u_t(0, x) = x$, $x \in \mathbb{R}$. Is it unique?